

Convergence Tests for Infinite Series

Answer Key

AP Calculus BC

Quick Answers

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|---------------------------------------|------------------------|
| 1. $\frac{3}{2}$ | 6. $\frac{11}{18}$ |
| 2. $\frac{1}{2}$ | 7. $\frac{1}{\log(2)}$ |
| 3. 1 | 8. $\frac{1}{1^{21}}$ |
| 4. $\frac{1549}{1728}, \frac{1}{125}$ | 9. $\frac{1}{5}$ |
| 5. 0 | 10. 0 |
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Curriculum

Level: AP Calculus BC

LIM-7, LIM-8 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 14 tagged checks.

1. $\sum_{n=0}^{\infty} \frac{1}{3^n}$: (a) find the exact sum; (b) use a direct comparison to explain why $\sum_{n=0}^{\infty} \frac{1}{3^n + n}$ converges.

Solution. (a) This is geometric with first term $a = 1$ (the $n = 0$ term) and ratio $r = \frac{1}{3}$. Since $|r| < 1$ it converges to $\frac{a}{1-r} = \frac{1}{1-\frac{1}{3}} = \frac{1}{\frac{2}{3}}$.

(b) For every $n \geq 0$ we have $3^n + n \geq 3^n > 0$, so $0 < \frac{1}{3^n + n} \leq \frac{1}{3^n}$. The larger series converges by (a), so by direct comparison the smaller one converges too. (Direct comparison needs the terms non-negative, which they are.)

$\frac{3}{2}$

2. Ratio test on $\sum_{n=1}^{\infty} \frac{n}{2^n}$.

Solution. With $a_n = \frac{n}{2^n}$,

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{n+1}{2^{n+1}} \cdot \frac{2^n}{n} = \frac{1}{2} \cdot \frac{n+1}{n}.$$

As $n \rightarrow \infty$, $\frac{n+1}{n} \rightarrow 1$, so $L = \frac{1}{2}$. Because $L < 1$ the series converges absolutely.

$L = \frac{1}{2} < 1$

3. Integral test on $\sum_{n=1}^{\infty} \frac{1}{n^2}$; evaluate $\int_1^{\infty} \frac{dx}{x^2}$.

Solution. On $[1, \infty)$, $f(x) = x^{-2}$ is positive, continuous, and $f'(x) = -2x^{-3} < 0$, so f is decreasing — the integral test applies. Then

$$\int_1^{\infty} x^{-2} dx = \lim_{b \rightarrow \infty} \left[-\frac{1}{x} \right]_1^b = \lim_{b \rightarrow \infty} \left(1 - \frac{1}{b} \right).$$

The improper integral converges, so the series converges. (The integral's value is not the sum of the series — it only decides convergence.)

1

4. $S = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$: (a) S_4 exactly; (b) the alternating-series error bound for $|S - S_4|$.

Solution. (a) The signs alternate starting positive:

$$S_4 = 1 - \frac{1}{8} + \frac{1}{27} - \frac{1}{64} = \frac{1728 - 216 + 64 - 27}{1728}.$$

(b) The alternating series test's remainder bound is the FIRST omitted term, $|a_5| = \frac{1}{5^3}$ — not the last one kept. The hypotheses hold: $\frac{1}{n^3}$ is positive, decreasing, and tends to 0.

$S_4 = \frac{1549}{1728}, \quad |S - S_4| \leq \frac{1}{125}$

5. Ratio test on $\sum_{n=0}^{\infty} \frac{2^n}{n!}$.

Solution. With $a_n = \frac{2^n}{n!}$,

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} = \frac{2}{n+1},$$

using $(n+1)! = (n+1) \cdot n!$. As $n \rightarrow \infty$ this tends to 0, and $L < 1$, so the series converges absolutely. (Factorials beat exponentials, which is why the ratio test is the right tool here.)

$L = 0$

6. $\sum_{n=1}^{\infty} \frac{1}{n^2 + 3n}$: (a) comparison; (b) exact sum by telescoping.

Solution. (a) For $n \geq 1$, $n^2 + 3n > n^2 > 0$, so $0 < \frac{1}{n^2 + 3n} < \frac{1}{n^2}$. The p -series with $p = 2$ converges, so by direct comparison this series converges.

(b) With the given split, the N th partial sum is

$$\frac{1}{3} \sum_{n=1}^N \left(\frac{1}{n} - \frac{1}{n+3} \right) = \frac{1}{3} \left(1 + \frac{1}{2} + \frac{1}{3} - \frac{1}{N+1} - \frac{1}{N+2} - \frac{1}{N+3} \right).$$

The three tail terms go to 0, leaving $\frac{1}{3} \cdot \frac{11}{6}$.

$\frac{11}{18}$

7. Integral test on $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$.

Solution. On $[2, \infty)$, $f(x) = \frac{1}{x(\ln x)^2}$ is positive, continuous and decreasing (both x and $\ln x$ increase), so the integral test applies. Substituting $u = \ln x$, $du = \frac{dx}{x}$:

$$\int_2^{\infty} \frac{dx}{x(\ln x)^2} = \int_{\ln 2}^{\infty} \frac{du}{u^2} = \lim_{b \rightarrow \infty} \left[-\frac{1}{u} \right]_{\ln 2}^b.$$

The limit is finite, so the improper integral converges and therefore so does the series.

$$\boxed{\frac{1}{\ln 2}}$$

8. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$: smallest N with $|S - S_N| < 0.01$, and the bound $|a_{N+1}|$.

Solution. The alternating-series bound is $|S - S_N| \leq |a_{N+1}| = \frac{1}{(N+1)^2}$. Require

$$\frac{1}{(N+1)^2} < \frac{1}{100} \iff (N+1)^2 > 100 \iff N+1 > 10.$$

The smallest integer is $N+1 = 11$, i.e. $N = 10$ terms, and then the guaranteed bound is $\frac{1}{11^2}$. (Check: $N = 9$ would give $\frac{1}{100}$, which is not strictly less than 0.01.)

$$\boxed{N = 10, \quad |a_{11}| = \frac{1}{121}}$$

9. Radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{n(x-2)^n}{5^n}$.

Solution. With $a_n = \frac{n(x-2)^n}{5^n}$,

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{n+1}{n} \cdot \frac{|x-2|}{5} \longrightarrow \frac{|x-2|}{5}.$$

The coefficient part $\frac{n+1}{5n}$ tends to $\frac{1}{5}$, so the ratio-test limit is $\frac{1}{5}|x-2|$. Convergence needs this < 1 , i.e. $|x-2| < 5$, so the radius is $R = 5$ and the open interval is $-3 < x < 7$. (Endpoints need separate tests and are not asked for here.)

$$\boxed{L = \frac{1}{5}|x-2|, \quad R = 5, \quad (-3, 7)}$$

10. Show $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n^2+1}$ converges but not absolutely.

Solution. Let $b_n = \frac{n}{n^2+1} > 0$.

Convergence (alternating series test). Dividing top and bottom by n^2 gives $b_n \rightarrow 0$. And b_n is eventually decreasing: treating n as continuous,

$$\frac{d}{dn} \left(\frac{n}{n^2+1} \right) = \frac{1-n^2}{(n^2+1)^2} < 0 \quad \text{for } n > 1.$$

Both hypotheses hold, so the alternating series converges.

Not absolute (direct comparison). For $n \geq 1$, $n^2 + 1 \leq 2n^2$, so $b_n = \frac{n}{n^2 + 1} \geq \frac{n}{2n^2} = \frac{1}{2n}$. The harmonic series diverges, so $\sum b_n$ diverges by direct comparison. Hence the series converges conditionally.

 $\lim b_n = 0$; conditionally convergent